

Ionically Conductive Elastic Polymer Binder for Ultrahigh Loading Electrode in High-Energy-Density Lithium Batteries

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Despite the increasing demand for high-energy-density lithium batteries, the development of high-mass-loading electrodes remains challenged by structural instability and poor charge transfer. Herein, an ionically conductive elastic polymer (ICEP) binder, designed to enable the fabrication of ultrahigh mass-loading Ni-rich layered cathodes ($\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$, NCM811), is introduced. The ICEP binder integrates mechanical elasticity, strong adhesion, and ionic conductivity through diverse functional groups, addressing challenges in high-mass-loading electrode fabrication. Hydrogen bonding between the ICEP binder and NCM811 particles ensures uniform electrode morphology, forming a stable cathode–electrolyte interphase (CEI). This stable interface mitigates surface side reactions, suppresses phase transitions in NCM811, and improves long-term electrochemical stability. Additionally, the ICEP binder enhances Li-ion diffusivity, reduces interphase resistance, and promotes faster electrochemical kinetics, while preventing solvent-drying-induced cracking. As a result, high-mass-loading electrodes (62.4 mg cm^{-2} , 12.5 mAh cm^{-2}) are successfully fabricated with the ICEP binder and demonstrate 94.6% capacity retention. Furthermore, a double-stacked pouch-type lithium metal full cell incorporating ICEP-based cathodes achieves energy densities of $377.6 \text{ Wh kg}_{\text{cell}}^{-1}$ and $1016.8 \text{ Wh L}_{\text{cell}}^{-1}$ (including package materials), setting new benchmarks for lithium metal batteries. These findings establish ICEP as a highly effective binder for next-generation high-energy-density batteries, offering a scalable and commercially viable solution for ultrahigh-loading cathodes.

1. Introduction

The growing demand for high-energy-density lithium (Li) batteries in applications such as electric vehicles and urban air mobility has driven intensive research into improving battery performance and scalability.^[1] A critical approach to enhancing energy density involves increasing electrode thickness, which reduces the weight fraction of non-active components such as separators and current collectors.^[2] However, fabricating thick electrodes presents inherent processing and mechanical challenges that hinder their practical implementation.^[3]

One of the primary obstacles in thick electrode fabrication arises from solvent evaporation during the drying process, which induces capillary-driven segregation of binder materials toward the electrode surface.^[4] This phenomenon weakens adhesion to the current collector, leading to crack formation and delamination. As a result, thick electrodes are significantly more structurally vulnerable than their conventional counterparts. Moreover, the extended Li-ion diffusion path in thick electrodes results in limited ion transport

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and sluggish through-thickness reaction kinetics.^[5] Several processing strategies, such as phase separation techniques^[6] and dry electrode fabrication,^[7] have been explored to mitigate polymer binder migration. However, these methods often introduce additional manufacturing complexities and cost constraints, impeding practical application. An alternative approach is to develop functional binders that strongly interact with active materials, thereby preventing segregation during electrode fabrication without altering the conventional electrode fabrication process.^[8]

Polyvinylidene fluoride (PVDF) remains the most widely adopted binder in cathode fabrication due to its high oxidative stability (≈ 4.7 V vs Li/Li⁺) and electrochemical inertness.^[9] However, its adhesion mechanism relies solely on weak van der Waals interactions, resulting in non-uniform binder coverage that leaves active materials exposed to electrolytes.^[10] This poor interfacial coverage leads to the formation of inhomogeneous and thick cathode–electrolyte interphase (CEI) layers, accelerating transition metal (TM) dissolution and promoting undesirable side reactions.^[11] Furthermore, in thick electrodes, PVDF exhibits uneven dispersion, and its rigid polymer backbone fails to accommodate both capillary-induced stresses during fabrication and volume changes during cycling. Beyond these material limitations, increasing global regulatory restrictions on per- and polyfluoroalkyl substances (PFAS) further complicate the long-term viability of PVDF-based binders.^[12] Therefore, the development of next-generation fluorine-free binders that offer superior adhesion, elasticity, and ionic conductivity is a pressing challenge.

An ideal polymer binder for high-mass-loading cathodes must exhibit high elasticity to withstand mechanical stresses during both electrode fabrication and repeated cycles. Additionally, it should incorporate functional groups capable of hydrogen bonding with the metal oxides on the surface of cathode materials, thereby enhancing interfacial adhesion and uniform particle coverage.^[13] This prevents direct exposure of active materials to the electrolyte, suppressing CEI overgrowth and surface side reactions that results in structural transformation. Furthermore, a rationally designed polymer binder should be capable of capturing dissolved TM ions from cathode materials to suppress the migration to the anode.^[8a] While previous binder development strategies have demonstrated improvements in structural stability and electrochemical performance, most fail to achieve simultaneous optimization of mechanical properties, interfacial chemistry, and ionic transport.^[9a,11,14] These limitations become particularly pronounced in thick electrodes exceeding 6 mAh cm^{−2}, where conventional binders struggle to maintain both electrochemical uniformity and structural cohesion. Thus, the design of copolymer binders with multiple functional groups that can synergistically address these challenges is essential for enabling high-energy-density Li batteries.

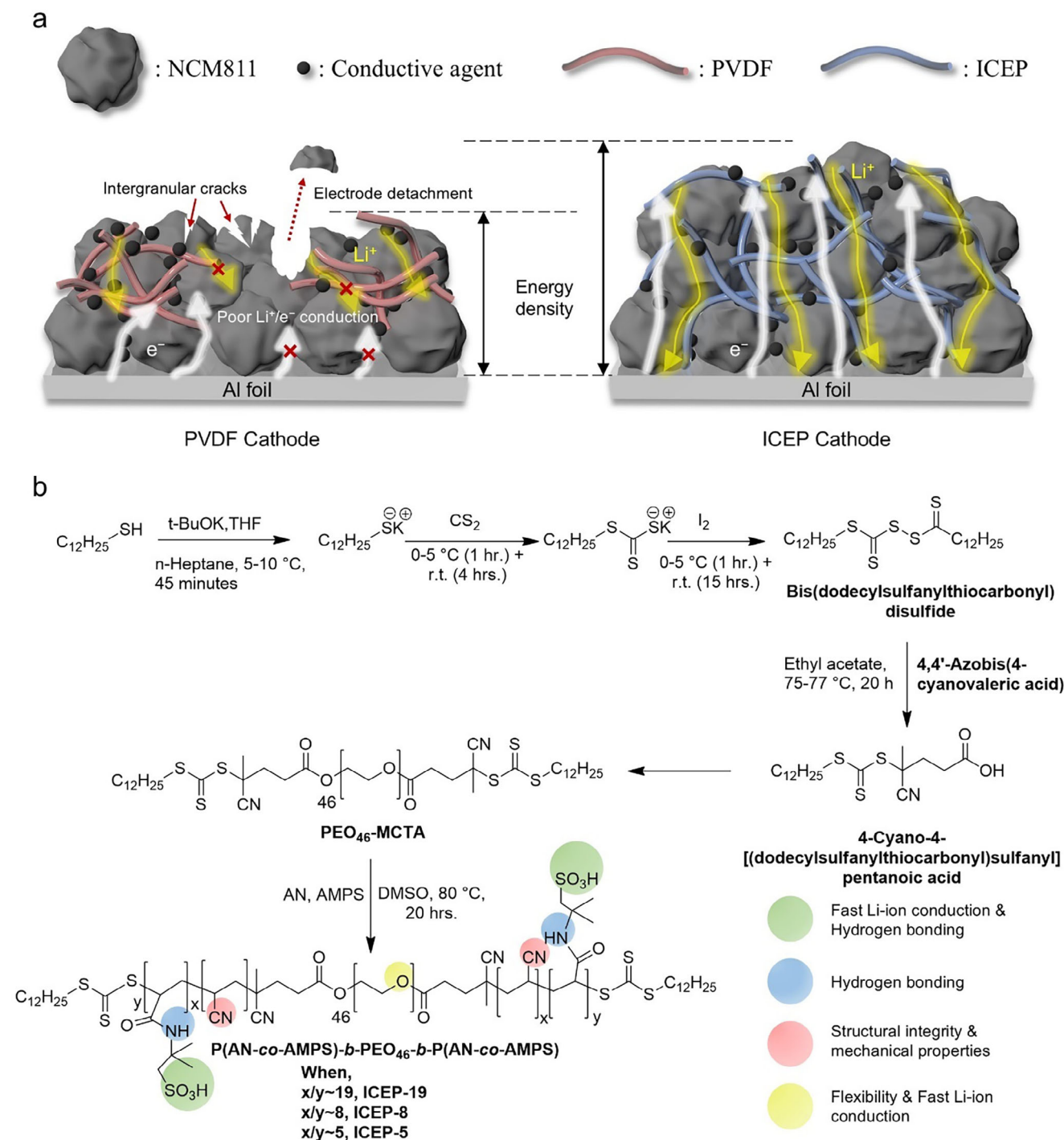
In this study, we introduce an ionically conductive elastic polymer (ICEP) binder that enables the fabrication of ultrahigh loading Ni-rich layered cathodes (LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂, NCM811) while simultaneously enhancing ion/electron conduction to increase the energy density of Li batteries (Scheme 1a). The ICEP binder exhibits superior elasticity and adhesion, effectively mitigating mechanical stress accumulation and preventing active material detachment and electrode cracking. Strong hydrogen bonding interactions between the ICEP binder and NCM811 particles further enhance electrode homogeneity, promoting the

formation of a thin, uniform CEI that suppresses electrolyte-induced surface side reactions. Moreover, the ICEP binder enhanced the Li-ion diffusivity and lowered the interphase/charge transfer resistance to induce faster electrochemical reaction kinetics compared to conventional cathodes with PVDF binder. Synchrotron-based nano-computed tomography (nano-CT) and scanning transmission electron microscopy (STEM) analyses confirmed that the ICEP binder significantly reduces the phase transformation and intergranular cracks of NCM811 over prolonged cycling. Notably, an ultrahigh loading of 62.4 mg cm^{−2}, corresponding to an areal capacity of 12.5 mAh cm^{−2}, was successfully achieved using ICEP, demonstrating exceptional cycling performance. Furthermore, a double-stacked pouch-type Li metal full cell incorporating ICEP-based cathodes paired with 100 μm Li metal, negative-to-positive capacity (N/P) and electrolyte-to-capacity (E/C) ratios of 1.57 and 2.5, respectively, achieved an outstanding energy density of 377.6 Wh kg_{cell}^{−1} and 1016.8 Wh L_{cell}^{−1} (including packaging materials), placing it among the highest reported values for Li metal pouch-type full cells. This study presents a scalable and effective binder design strategy for next-generation Li batteries, offering a practical pathway toward high-energy-density Li batteries with ultrahigh mass loading, long-term cycling stability, and superior interfacial protection.

2. Results and Discussion

2.1. Synthesis and Characterization of ICEP Binder

The ICEP binder is a block copolymer comprising of acrylonitrile (AN), 2-acrylamido-2-methylpropane sulfonic acid (AMPS), and ethylene oxide (EO) units, forming the structure ([P(AN-co-AMPS)]₂-b-PEO). This design concept draws inspiration from classical elastomers such as PS-b-PI-b-PS or PS-b-PB-b-PS (where PS = polystyrene, PI = polyisoprene, and PB = polybutadiene), but incorporates ionic functionalities, including ether units and sulfonic acids, to facilitate Li-ion transport, thereby rendering the material highly suitable as a multifunctional polymer binder for Li batteries (Scheme 1b). Specifically, the AN unit provides mechanical robustness and strong affinity with active materials through its highly polar C≡N group,^[15] while the AMPS unit improves ionic conductivity via its sulfonate (−SO₃H) groups and forms hydrogen bonds with cathode materials.^[16] The EO segment enhances polymer flexibility and promotes ionic transport through the ether oxygen groups along the PEO chain.^[17] A fundamental advantage of the ICEP architecture is the ability to fine-tune properties by varying the AN/AMPS ratio. Increasing the AN fraction enhances rigidity and mechanical strength, but may diminish elasticity and ionic conductivity. Conversely, increasing the AMPS content improves ionic transport and softens the polymer, but may compromise mechanical stability. Thus, an optimal balance of AN and AMPS is necessary to simultaneously achieve robust mechanical integrity, high ionic conductivity, and (electro)chemical stability, including insolubility in organic carbonate electrolytes. The synergy of these components enables the ICEP binder to facilitate Li-ion coordination and transport within the polymer matrix. The synthesis of the ICEP was carried out using a reversible addition-fragmentation chain transfer (RAFT) polymerization technique, utilizing a bifunctional PEO-macro chain



transfer agent (PEO-MCTA) prepared through esterification of commercial poly(ethylene glycol) (PEG) with ω - and α -terminated hydroxyl groups and a carboxylic acid-functional RAFT agent (see the details of the synthesis process in Experimental Section). This process enabled the symmetric growth of P(AN-co-AMPS) random copolymer chains on both sides of the PEO-MCTA, ensur-

ing precise control over the polymer composition and architecture (Figure S1, Supporting Information).

Three different composition ratios, ICEP-5, ICEP-8, and ICEP-19, were synthesized by varying the [AN]/[AMPS] ratio, which was carefully optimized to balance the mechanical robustness and ionic conductivity of the polymer (Table S1, Supporting

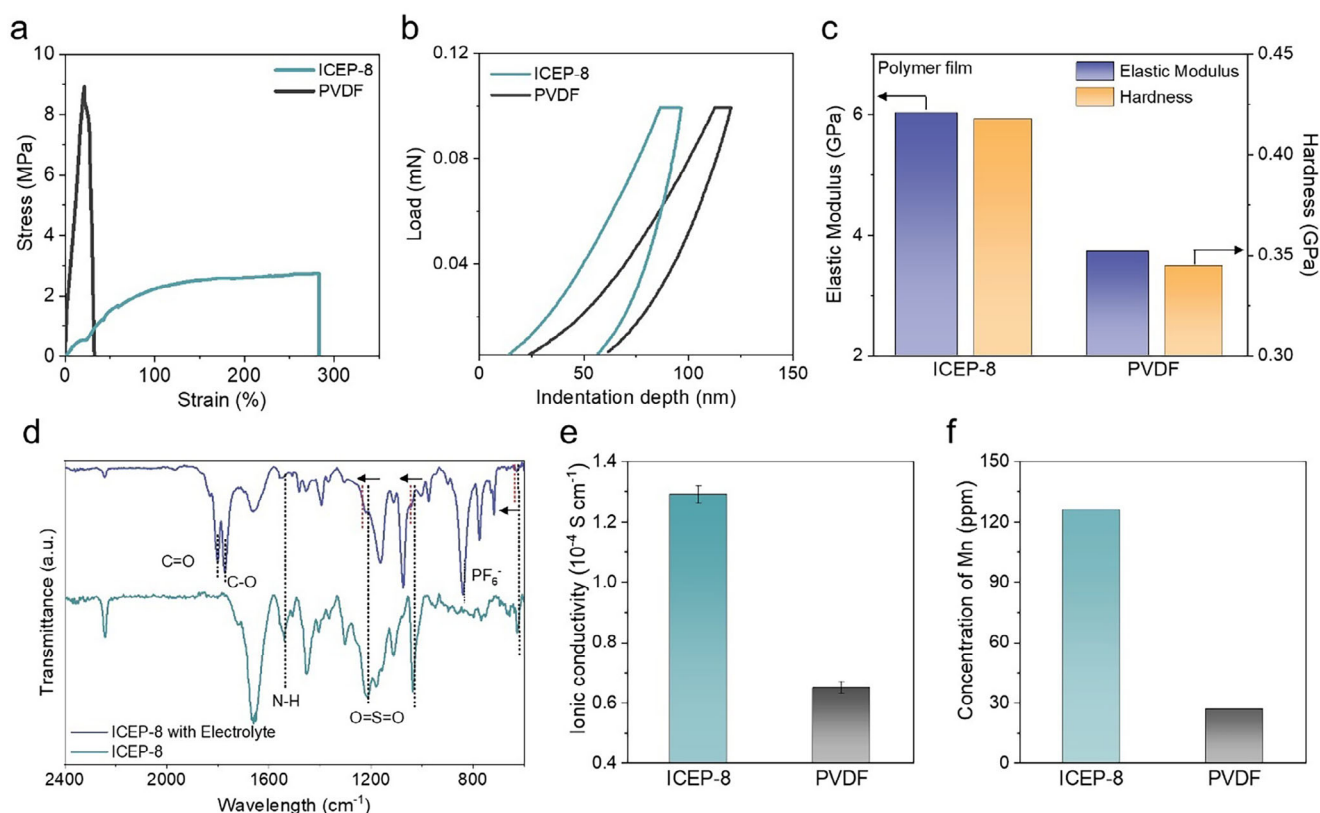


Figure 1. Chemomechanical and ion-conductive characteristics of the ICEP binder. a) Tensile S-S curves of ICEP-8 and PVDF polymer films. b) Load-displacement profiles obtained from nanoindentation tests on ICEP-8 and PVDF polymer films and corresponding elastic modulus and hardness values. d) FT-IR spectra of pristine and electrolyte-immersed ICEP-8 films. e) Ionic conductivity of ICEP-8 and PVDF polymers. f) Quantitative analysis of Mn²⁺ ions captured by ICEP-8 and PVDF polymer films.

Information). The degree of polymerization for the EO unit was fixed at 46 using PEO₄₆-MCTA. The molecular weights of the resulting copolymers were ~100 kDa, as confirmed by gel permeation chromatography (GPC), and ¹H NMR analysis confirmed the composition ratios (Figures S2 and S3, Supporting Information).

Differential scanning calorimetry (DSC) analysis reveals the thermal transitions of these ICEP triblock copolymers and highlights how their structure–property relationships can be effectively tuned by varying the [AN]/[AMPS] ratio (Figure S4, Supporting Information). All ICEP samples displayed two glass transition temperatures (*T*_g): one below room temperature for the PEO midblock, imparting elasticity, and another above room temperature for the P(AN-co-AMPS) side blocks, functioning as physical crosslinks. Notably, ICEP-8, with an intermediate [AN]/[AMPS] ratio, exhibited the lowest *T*_g for both blocks (−56 °C for PEO; 85.2 °C for P(AN-co-AMPS)) and complete suppression of PEO crystallinity, indicating a highly amorphous, flexible network. In contrast, ICEP-5 and ICEP-19 showed higher PEO *T*_gs (−18 °C and −25 °C, respectively) and partial crystallinity, suggesting more rigid phase morphologies and limited chain mobility. Atomic force microscopy (AFM) phase imaging supports these thermal transitions. ICEP-8 displayed well-defined, phase-separated domains (Figure S5, Supporting Information), indicating sufficient interfacial mobility and softness to

support elastic behavior, enabling stress dissipation through a co-continuous soft phase network. In contrast, ICEP-19 showed more homogeneous and potentially stiffer morphology, while ICEP-5 exhibited less regular morphology, possibly due to defective or poorly connected phase separation.

The mechanical properties of the ICEP binder were analyzed through tensile stress–strain (S-S) curves obtained from free-standing films. The ICEP-8 binder exhibited an impressive elongation at break of 283% and a toughness of 601.2 J m^{−3}, surpassing the performance of the conventional PVDF binder, which showed an elongation at break of only 31.8% and a toughness of 151.8 J m^{−3} (Figure 1a). These results highlight the remarkable elasticity and energy absorption capacity of the ICEP-8 binder, which enables it to deform plastically without fracturing, a key feature for alleviating cracks caused by drying-induced capillary stresses. Due to the well-phase-separated morphology (with soft and hard blocks distinct yet continuous), fully amorphous and flexible middle block, and balanced AN/AMPS ratio providing rigidity and ionic interaction without excessive stiffness, ICEP binder exhibited the highest elongation-at-break and toughness. Further analysis of different ICEP variants with varying [AN]/[AMPS] ratios demonstrated that ICEP-5, although exhibiting slightly higher elongation at break (212%) than ICEP-19 (167%), had lower toughness (290.8 J m^{−3}) compared to ICEP-19 (481 J m^{−3}; Figure S6, Supporting Information). This reduction

in toughness for ICEP-5 is likely due to enhanced ionic clustering, which stiffens the matrix, and defective or poorly connected phase separation, which limits effective stress dissipation. In contrast, the higher proportion of rigid AN in ICEP-19 contributes to increased toughness but results in lower elongation at break due to the inherently glassy nature of AN. Nanoindentation test provided further evidence of the superior mechanical properties of the binder. Under a constant load of 0.1 mN, the ICEP-8 binder exhibited a smaller indentation depth of 96.5 nm compared to that of the PVDF binder (120.0 nm), suggesting that the ICEP-8 binder has superior mechanical resistance (Figure 1b). Moreover, the ICEP-8 binder achieved an elastic modulus of 6.03 GPa and a hardness of 0.42 GPa in the nanoindentation test, significantly exceeding the values of 3.73 GPa and 0.34 GPa observed for the PVDF binder (Figure 1c). The enhanced mechanical performance is attributed to the synergistic contributions of the P(AN-co-AMPS) side blocks, which feature a balanced ratio of rigid AN and ionic AMPS units, and the flexible middle PEO block. Given the superior toughness, mechanical elasticity, and amorphous nature confirmed by DSC, ICEP-8 was selected as the representative binder formulation for further investigation throughout this study.

The interaction between the ICEP-8 binder and battery electrolytes was investigated using Fourier transform infrared (FT-IR) spectroscopy (Figure 1d and Figure S7, Supporting Information). After immersion of the ICEP-8 binder film in the electrolyte, new vibration peaks were observed at 1803 cm^{-1} and 1774 cm^{-1} , corresponding to the $\text{C}=\text{O}$ and $\text{C}-\text{O}$ groups from carbonate solvents. A peak at 839 cm^{-1} , associated with the $\text{P}-\text{F}$ bond of PF_6^- , was also detected.^[19] Additionally, the $\text{N}-\text{H}$ bending vibration frequency ($\approx 1540 \text{ cm}^{-1}$) of the AMPS unit shifted slightly higher wavenumber, suggesting hydrogen bonding interactions with electrolyte components. The characteristic vibrations of the $\text{O}=\text{S}=\text{O}$ group in the SO_3H units shifted from 1215 cm^{-1} and 1037 cm^{-1} to 1220 cm^{-1} and 1042 cm^{-1} , respectively. Interestingly, the $\text{C}-\text{S}$ peak, initially at 626 cm^{-1} , shifted to 629 cm^{-1} in the electrolyte-immersed film, indicating moderate interaction between Li^+ and SO_3H groups of AMPS.^[20] To further verify this interaction at the atomic scale, ^7Li NMR spectroscopy was conducted (Figure S8, Supporting Information). In deuterated DMSO, the addition of ICEP-8 to a LiPF_6 solution resulted in a downfield shift of the ^7Li resonance from 0.00 ppm (LiPF_6 only) to +0.05 ppm. This shift directly indicates coordination between Li^+ ions and the sulfonate groups in ICEP-8.^[21] This interaction is believed to facilitate Li -ion conduction within the ICEP-8 binder, which is critical for enhancing the performance of thick electrodes. To quantify the ionic conductivity of the ICEP-8 binder, electrochemical impedance spectroscopy (EIS) measurements were performed with polymer films placed between stainless steel electrodes. The results revealed that the ICEP-8 binder exhibited an ionic conductivity of $1.35 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature, more than double that of the conventional PVDF binder, which showed a conductivity of $0.65 \times 10^{-4} \text{ S cm}^{-1}$ (Figure 1e and Figure S9, Supporting Information). This improved ionic conductivity stems from the sulfonic acid groups in AMPS and the oxygen atoms in EO, which facilitate the movement of solvated ions along the polymer chains.^[22]

Another critical challenge in high-capacity and high-voltage cathodes, such as NCM811, is the dissolution of TM ions into

the electrolyte, which can lead to interfacial side reactions and metallic deposition on the anode. To address this issue, the TM-ion chelating ability of the ICEP-8 binder was investigated by immersing its films in a model liquid electrolyte containing 0.01 M manganese perchlorate [$\text{Mn}(\text{ClO}_4)_2$] for 6 h. The amount of trapped Mn^{2+} was then quantified using inductively coupled plasma mass spectrometry (ICP-MS). The results showed that the ICEP-8 binder retained 126 ppm of Mn^{2+} , significantly higher than the 27 ppm retained by the PVDF binder (Figure 1f). This enhanced chelating capability is attributed to the polar functional groups in the ICEP, such as acrylamide and sulfonic acid, effectively preventing TM ion dissolution and mitigating undesirable side reactions during battery cycling.

2.2. Multiscale Structural and Mechanical Characterization of ICEP Cathodes

Building on the exceptional chemomechanical properties of the ICEP-8, it was employed to fabricate NCM811 cathodes. To assess binder distribution and electrode homogeneity, analyses were conducted at the nanometer, micrometer, and millimeter scales. At the nanometer scale, transmission electron microscopy (TEM) revealed that the ICEP-8 binder formed a uniform and thin coating layer on the NCM811 particles, whereas the PVDF binder exhibited uneven, sporadic aggregation on particle surfaces (Figure 2a,d). The superior uniformity of the ICEP-8 binder was attributed to the strong hydrogen bonding interactions between its functional groups and the oxygen-containing groups on NCM811 particles, whereas the PVDF binder only relies on weak van der Waals forces. This robust interaction facilitated tight adhesion and elastic adaptability, ensuring comprehensive particle coverage during slurry preparation and electrode fabrication. Importantly, the ICEP-8 layer was sufficiently thin ($\approx 7 \text{ nm}$) to maintain effective Li -ion access and avoid kinetic hindrance to charge transfer reactions at the cathode surface. At the micrometer scale, scanning electron microscopy (SEM) indicated that the NCM811 cathodes with the ICEP-8 binder (denoted as ICEP-8 cathodes) exhibited uniformly distributed particles throughout the electrode. In comparison, the cathodes with PVDF binder (denoted as PVDF cathodes) showed significant particle aggregation and inhomogeneity (Figure 2b,e). To further evaluate the morphological homogeneity in the through-thickness direction, cross-sectional SEM analysis was performed on an ion-milled thick ICEP-8 cathode ($\approx 230 \text{ }\mu\text{m}$) and PVDF ($\approx 190 \text{ }\mu\text{m}$) cathodes. SEM images at different locations of the ICEP-8 cathode revealed a highly uniform dispersion of NCM811 particles and conductive agents throughout the electrode depth without noticeable aggregation of phase separation (Figure S10a, Supporting Information). Conversely, the PVDF cathode displayed pronounced thickness variations ($\approx 7.3 \text{ }\mu\text{m}$ difference across measured sites) and localized aggregation of conductive agents and binder, reflecting inferior dispersion and morphological homogeneity (Figure S10b, Supporting Information). Moving to the millimeter scale, three-dimensional (3D) laser confocal microscopy displayed markedly lower height deviation across a $1.0 \text{ mm} \times 1.0 \text{ mm}$ area, indicating a smoother and more uniform surface than PVDF cathodes (Figure 2c,f). This enhanced surface uniformity, attributed to the superior dispersion properties of the ICEP-8 binder,

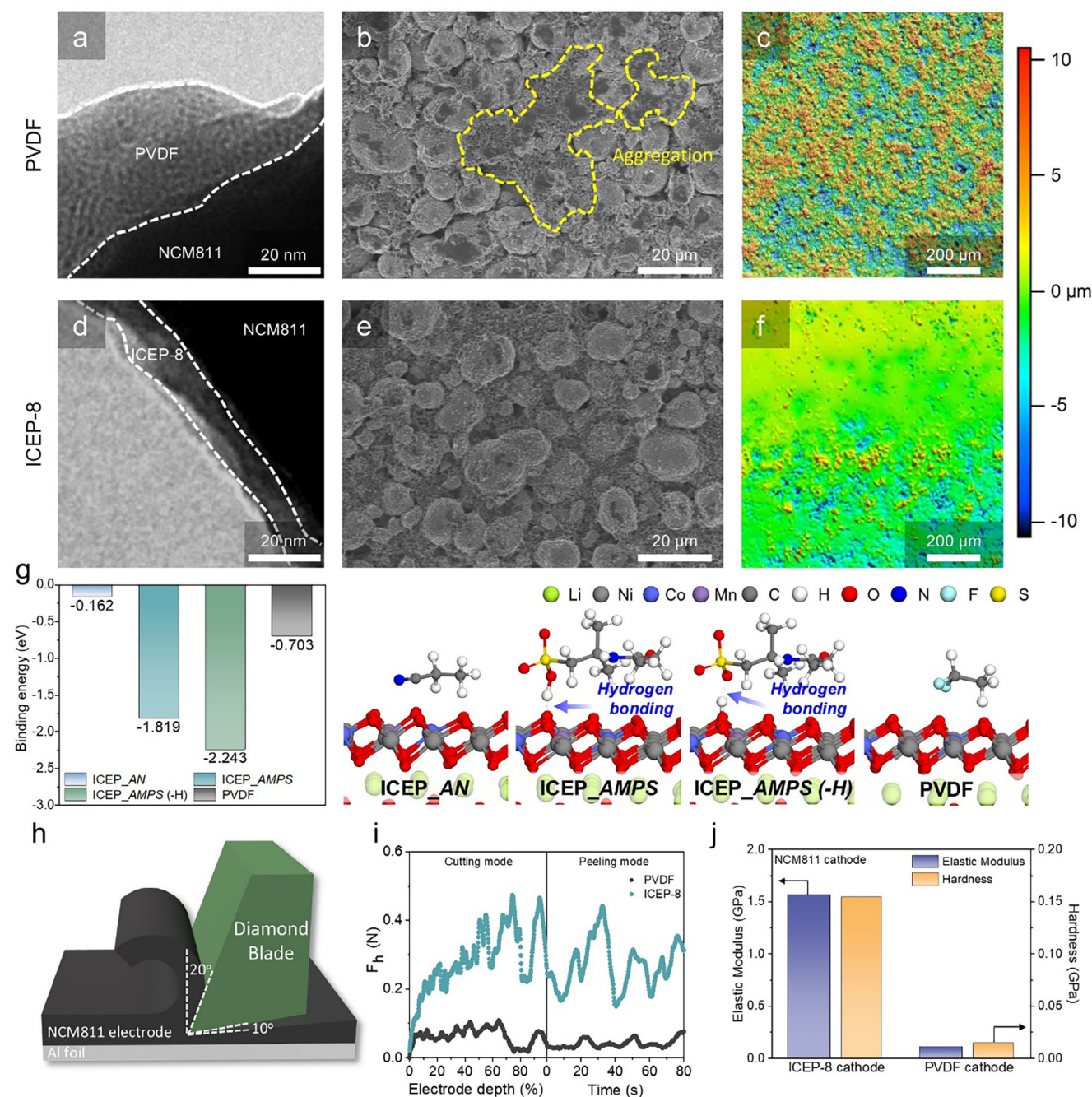


Figure 2. TEM images of a) PVDF and d) ICEP-8 on the NCM811 surface in the pristine state. Top SEM images of the b) PVDF and e) ICEP-8 cathodes. 3D Optical Profiler images of the cathodes with c) PVDF and f) ICEP-8 binders. g) Binding energies and optimized structures of ICEP_AN, ICEP_AMPS, ICEP_AMPS (-H), and PVDF on NCM811 (001) surface. h) The schematic illustration of the SAICAS experiment. i) The adhesive strength of the cathodes with PVDF and ICEP-8 binders at various electrode depths. j) Elastic modulus and hardness of NCM811 cathodes with different binders.

directly influenced the ionic/electrical conductivity of the NCM811 cathodes. Improved dispersion within the slurry resulted in well-developed electron conduction networks, giving the ICEP-8 cathode higher electrical conductivity compared to the PVDF cathode at the same composition ratio (Figure S11, Supporting Information).^[8a,23]

To further elucidate the molecular origins of this interfacial behavior, we conducted density functional theory (DFT) calcula-

tions to investigate the binding interactions between model segments of the ICEP and PVDF polymers and the NCM811 surface (Figure S12, Supporting Information). The ICEP structure was dissected into two representative functional blocks, denoted as ICEP_AN and ICEP_AMPS. Notably, ICEP_AMPS exhibited a significantly stronger binding affinity to the NCM811 surface compared to PVDF, primarily due to robust hydrogen bonding between the sulfonate group and surface oxygen atoms (Figure 2g

and Figure S13, Supporting Information). These DFT calculation results provide direct theoretical support for the experimentally observed interfacial adhesion and underpin the superior cohesion achieved with ICEP-8 binder.

The adhesion and cohesion properties of the ICEP-8 binder were evaluated using a surface and interfacial cutting analysis system (SAICAS) with a diamond blade (width: 1 mm) under a shear angle of 45°, rake angle of 20°, and clearance angle of 10° (Figure 2h).^[24] In cutting mode, the cohesion of binder is measured by how well the material can resist shear forces during electrode slicing, which is essential for maintaining the structural integrity of high-mass-loading electrodes and preventing delamination. In peeling mode, the adhesive strength of binder is evaluated by the force required to separate the electrode from the current collector. This measures the tensile resistance of the binder under separation stress, which is crucial for ensuring that the cathode remains intact during long-term cycling. The ICEP-8 cathodes demonstrated significantly higher cohesion and adhesion strength in both cutting and peeling modes compared to PVDF cathodes (Figure 2i). Specifically, the ICEP-8 cathodes exhibited average cohesion and adhesion forces of 0.29 N and 0.27 N, respectively, while the PVDF cathodes displayed forces of only 0.07 N and 0.04 N. This enhanced performance is attributed to the strong hydrogen bonding facilitated by the polar functional groups in the ICEP-8 binder. These interactions are critical for maintaining the integrity of high-mass-loading electrodes, ensuring uniform redox behavior and electrochemical stability over prolonged cycles.

Mechanical stability of the ICEP-8 cathodes was further verified through electrode nanoindentation tests conducted at a higher load (2.0 mN) compared to the polymer film nanoindentation in Figure 1c (0.1 mN) to account for the presence of inorganic particles (NCM811) and conductive additives. Load-displacement curves revealed that the ICEP-8 cathode exhibited significantly smaller indentation depths (936.1 nm) than PVDF cathodes (3345.2 nm), highlighting their superior mechanical strength (Figure S14, Supporting Information). Moreover, the ICEP-8 cathodes achieved a higher elastic recovery rate (48.0%) compared to the PVDF cathodes (35.8%), demonstrating their greater ability to withstand mechanical stress with minimal plastic deformation. Additionally, the elastic modulus and hardness of the NCM811 electrodes were consistent with these findings. As shown in Figure 2j, the ICEP-8 cathodes displayed an elastic modulus of 1.57 GPa and a hardness of 0.15 GPa, both exceeding the values for PVDF cathodes (0.11 GPa and 0.014 GPa, respectively) by more than an order of magnitude. This substantial enhancement in mechanical properties is attributed to the improved uniformity and elastic nature of the ICEP-8 binder within the electrode structure.

2.3. Electrochemical Performance and Li-Ion Transport Kinetics of ICEP Cathodes

Prior to battery evaluation, linear sweep voltammetry was conducted for the binders from 3.0 to 6.0 V (vs Li/Li⁺) at a scan rate of 1 mV s⁻¹ (Figure S15, Supporting Information). Both ICEP-8 and PVDF binders exhibited an onset of oxidative current increase above 5.0 V. Given that the operational voltage range of

NCM811 cathodes is confined between 2.7 and 4.2 V, the ICEP-8 binder ensures sufficient electrochemical stability during battery operation. The electrochemical characteristics of NCM811 cathodes were assessed in coin-type half cells with a high mass loading of 18.8 mg cm⁻². The ICEP-8 cathode demonstrated a slightly higher initial discharge capacity of 193.8 mAh g⁻¹ and an initial Coulombic efficiency of 90.8%, surpassing the values from the PVDF cathode (181.9 mAh g⁻¹ and 86.2%) (Figure S16, Supporting Information). Furthermore, the ICEP-8 cathode exhibited a significantly lower charge overpotential during the first cycle, reaching 3.77 V, compared to 3.89 V for the PVDF system. These results are primarily ascribed to the enhanced ionic conductivity and homogeneous particle distribution enabled by the strong hydrogen bonding interactions within the ICEP-8 binder. The long-term cycling stability of both cathodes was evaluated at a charge/discharge rate of 0.5 C (Figure 3a). After 170 cycles, the ICEP-8 cathode retained 85.5% of its initial capacity, whereas the PVDF cathode displayed rapid capacity degradation, retaining only 28.9% after 80 cycles. Before the onset of degradation, the average CE values over the first 70 cycles were 99.8% for ICEP-8 and 99.2% for PVDF. The lower CE of the PVDF cathode reflects continuous parasitic reactions at the cathode–electrolyte interface, such as transition metal dissolution and CEI thickening, stemming from poor interfacial coverage and inadequate ion transport. In contrast, the multifunctional ICEP-8 binder minimizes side reactions and promotes stable Li-ion conduction, leading to higher CE and enhanced cycling durability. To compare the rate performance, discharge capacities were recorded across multiple current densities, ranging from 0.2 C to 2.0 C, while maintaining a constant charge rate of 0.2 C (Figure 3b). The ICEP-8 cathode exhibited higher capacity retention across all rates, indicative of enhanced electrochemical kinetics and superior Li-ion conduction pathways compared to its PVDF counterpart. Furthermore, high C-rate cycling tests (2.0 C/2.0 C) were performed to evaluate rate-dependent degradation (Figure S17, Supporting Information). The PVDF cathode exhibited a low initial capacity (~99 mAh g⁻¹), rapidly fading to 39.4% retention after 40 cycles. By comparison, the ICEP-8 cathode delivered a higher initial capacity (153 mAh g⁻¹) and exhibited remarkable capacity retention of 90.0% after 120 cycles, highlighting its exceptional performance under high-rate operation.

To further investigate Li-ion transport properties, galvanostatic intermittent titration technique (GITT) measurements were conducted after 30 cycles (Figure S18, Supporting Information). The ICEP-8 cathode exhibited lower internal resistance (R_{internal}) throughout both charge and discharge processes, averaging 31.0 Ω, in contrast to the 57.2 Ω observed for the PVDF cathode (Figure 3c). Additionally, the Li-ion diffusion coefficient (D_{Li^+}) remained consistently higher in the ICEP system over the entire state-of-charge (SOC) and depth-of-discharge (DOD) ranges. The average D_{Li^+} value for the ICEP-8 cathode reached 0.42×10^{-7} cm² s⁻¹, more than twice that of the PVDF cathode (0.18×10^{-7} cm² s⁻¹).

The superior Li-ion transport kinetics and electrochemical performance of the ICEP-8 cathode were further supported by the distribution of relaxation time (DRT) analysis conducted on Li||NCM811 cells.^[25] In this analysis, charge transfer resistance (R_{ct}) at low frequencies ($10^{-2} < \tau < 10^0$) is attributed to Faradic process within the electrode, while the intermediate frequency

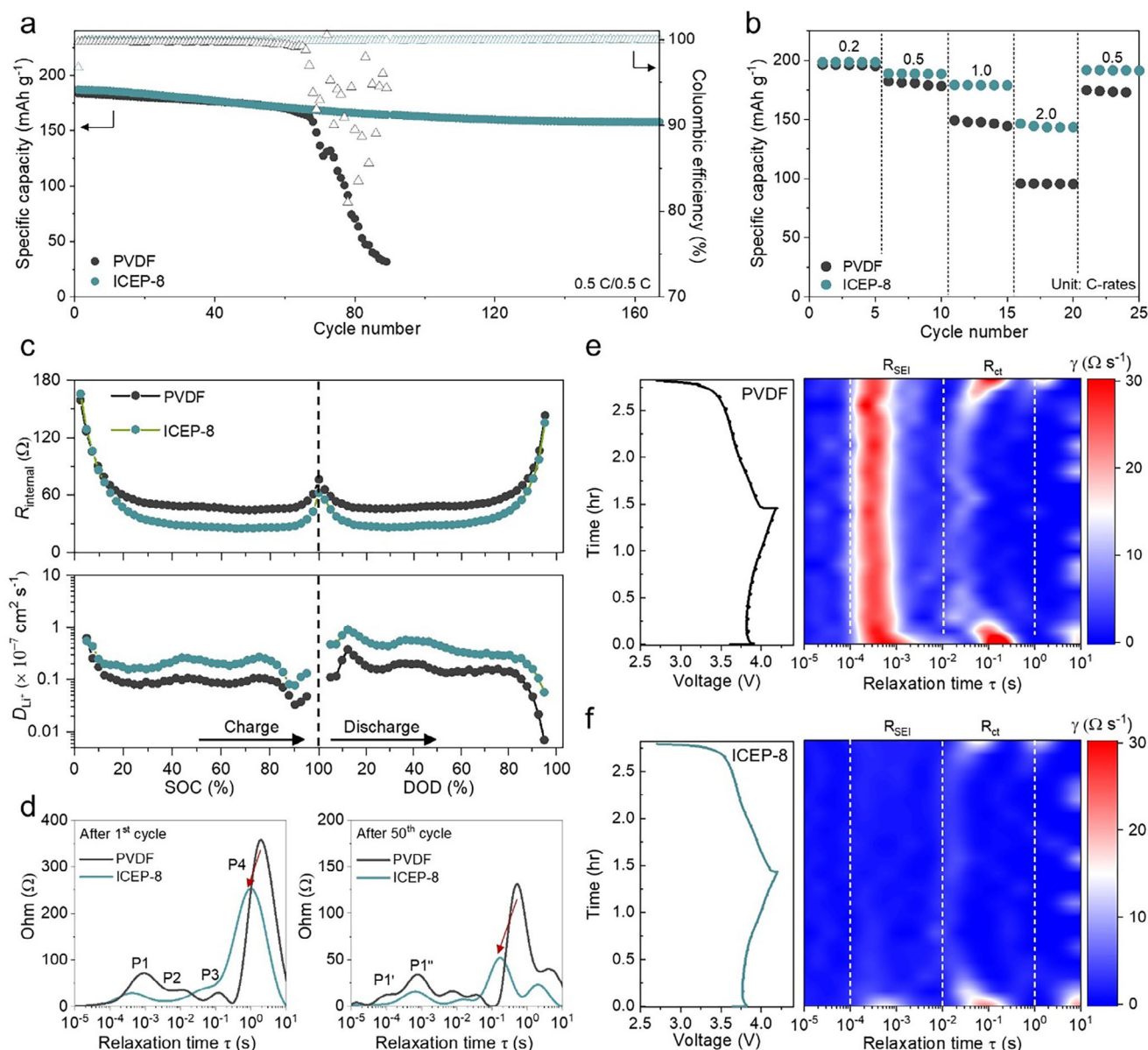


Figure 3. Electrochemical performance and Li-ion transport properties of NCM811 cathodes. a) Long-term cycling stability at 0.5 C and b) discharge rate capabilities at varying current densities of cathodes with PVDF and ICEP-8 binders. c) Internal resistance and Li-ion diffusion coefficients (D_{Li^+}) across different SOC and DOD. d) Ex situ DRT results of Li||NCM811 cells employing different binders after the first and 50th cycles. In situ DRT results of Li||NCM811 cells during cycling with e) PVDF and f) ICEP-8 binders.

peak ($10^{-2} < \tau < 10^{-4}$) corresponds to SEI resistance (R_{SEI}). After the first cycle, four distinct peaks (P1–P4) were identified in both cathodes, corresponding to interfacial reactions (P1), interfacial + charge transfer processes (P2), charge transfer reactions (P3), and charge transfer + solid-state diffusion (P4). Notably, the ICEP-8 cathode exhibited consistently lower resistance values across all regions and cycles compared to the PVDF cathode (Figure 3d). After 50 cycles, the total resistance in both cathodes decreased, attributed to CEI stabilization, which fosters more uniform Li-ion conduction. A notable difference between the two binder systems was the peak splitting observed in the PVDF cathode for P1, which is associated with R_{SEI} . This peak separation

suggests the formation of multiple resistive elements within the SEI layer, likely due to its heterogeneous and unstable growth. In contrast, the ICEP-8 cathode maintained a single, well-defined R_{SEI} peak, indicating a more stable and homogeneously grown interfacial layer. Additionally, the relaxation times of all resistive processes (R_{SEI} , R_{ct} , and solid-state diffusion) were significantly shorter for the ICEP-8 cathode than for the PVDF cathode, implying faster electrochemical reaction kinetics and efficient Li-ion transport within the ICEP system. To gain further insights into the evolution of interfacial resistance over cycling, in situ galvanostatic electrochemical impedance spectroscopy (GEIS)-combined DRT mapping was conducted (Figure 3e,f).^[25–26] The

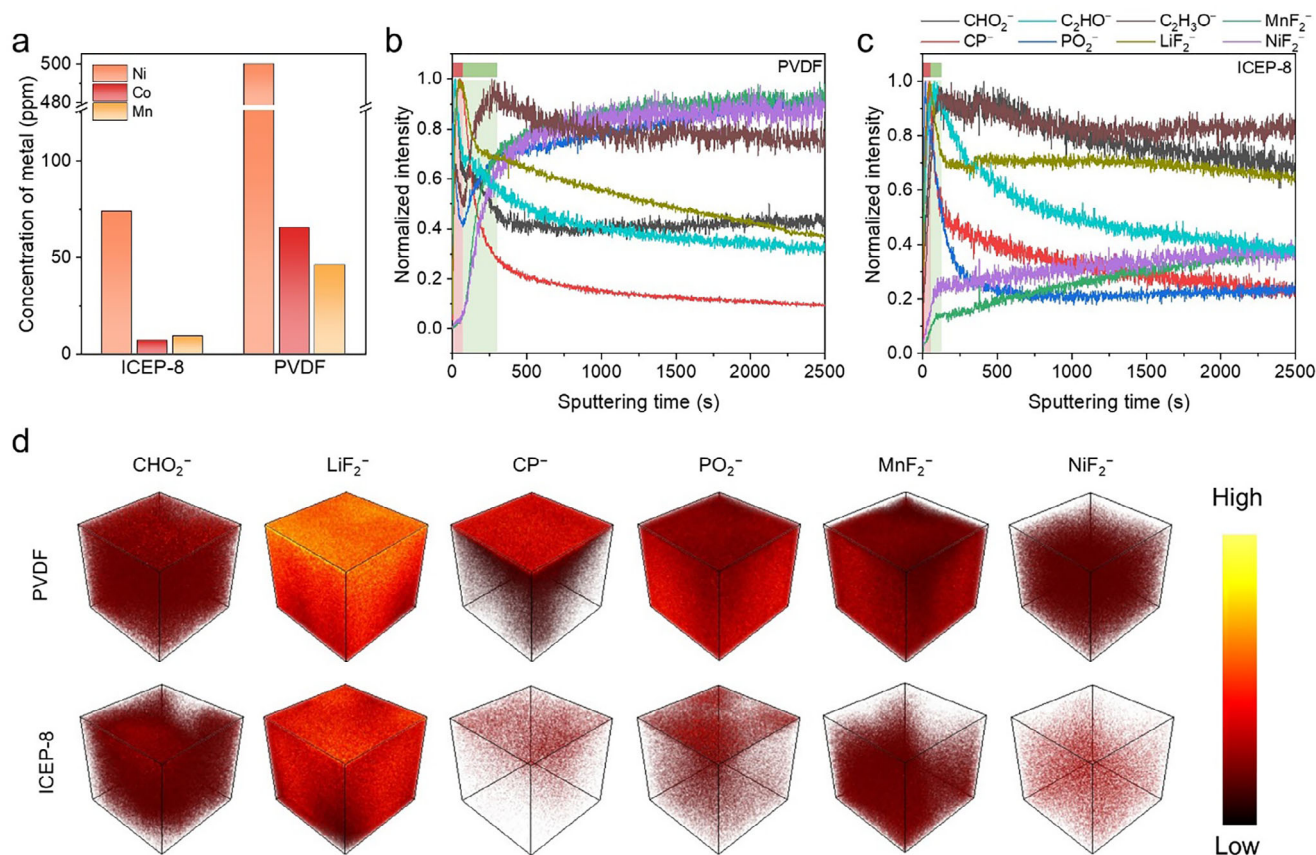


Figure 4. Interfacial stability analysis of NCM811 cathodes. a) Quantitative analysis of metallic Ni, Co, and Mn deposited on Li-metal anodes after cycling. Normalized TOF-SIMS depth profiles of representative secondary-ion fragments at the cathode-electrolyte interfaces formed on cycled NCM811 cathodes employing b) PVDF and c) ICEP-8 binders. d) 3D TOF-SIMS visualization of secondary-ion fragments on the cathode surface with different binders.

high R_{ct} values observed at low SOC and high DOD are attributed to the low mobility of Li-ions in the highly lithiated NCM811 phase, which is limited by fewer vacancies (Figure S19, Supporting Information). However, as the vacancies increase during cycling, the charge transfer resistance decreases, which is consistent with results from the GITT analysis. The ICEP-8 cathode exhibited negligible R_{ct} and R_{SEI} values throughout cycling, signifying faster and more uniform Li-ion conduction from the SEI layer to charge transfer processes within the electrode. In contrast, the PVDF cathode suffered from progressive resistance buildup, likely contributing to its rapid capacity decay and limited rate performance.

2.4. Structural and Interfacial Stability of ICEP Cathodes

The superior cycling stability and interfacial robustness of the ICEP-8 cathode were further confirmed using a series of post-mortem analyses. ICP-MS measurements revealed substantially lower concentrations of transition metals (Ni, Co, and Mn) deposited on the Li metal anode in cells cycled with the ICEP-8 cathode compared to those using the PVDF cathode (Figure 4a). While the presence of TM species on the Li metal surface (73.9 ppm Ni, 7.1 ppm Co, and 9.2 ppm Mn) confirms that some dis-

solution and transport still occur, the significantly lower levels than in the PVDF system suggest that the ICEP-8 binder contributes to both interfacial stabilization and partial chelation of dissolved TM ions. The interfacial characteristics of the cathodes were further examined using time-of-flight secondary-ion mass spectrometry (TOF-SIMS). Normalized depth profiles of key secondary-ion species associated with the CEI were analyzed, including organic fragments from electrolyte solvent decomposition (CHO_2^- , C_2HO^- , and $\text{C}_2\text{H}_3\text{O}^-$), inorganic species from salt or additive decomposition (CP^- , PO_2^- , and LiF_2^-), and TM fluoride (i.e., MnF_2^- and NiF_2^- ; Figure 4b,c).^[27] Both cathodes exhibited a distinct double-layer CEI structure: the outer layer (highlighted in red bar, ≈ 0 –50 s sputtering time for PVDF and 0–43 s for ICEP-8) was enriched in metal-derived fragments, while the inner CEI layer (highlighted in green bar) was composed primarily of electrolyte decomposition species. Notably, the ICEP-8 cathode showed a significantly shorter total sputtering time for the inner CEI region (up to 108 s), compared to the PVDF cathode (up to 283 s), indicating a thinner and more compact interphase. This suggests that the ICEP-8 binder suppresses excessive electrolyte decomposition and enables the formation of a more stable CEI during the repeated cycles. Further insights were gained through 3D depth reconstruction images of various CEI fragments (Figure 4d). The PVDF cathode exhibited

a higher intensity of solvent decomposition product (CHO_2^-) and a significantly greater presence of lithium salt and additive decomposition fragments (LiF_2^- , CP_2^- , and PO_2^-), reinforcing the observation of excessive electrolyte degradation and CEI overgrowth. Additionally, higher concentrations of MnF_2^- and NiF_2^- were detected in the PVDF cathode, indicating more severe structural deterioration due to acidic species such as HF attacking the cathode surface. Notably, the MnF_2^- signal was stronger than NiF_2^- despite higher Ni dissolution, as confirmed by ICEP-MS (Figure 4a). This reflects the greater tendency to form fluorides and higher ionization efficiency in TOF-SIMS, whereas Ni more readily forms oxides like NiO, resulting in weaker fluoride signals.^[28] X-ray photoelectron spectroscopy (XPS) analysis further supported the TOF-SIMS findings. The ICEP-8 cathode exhibited lower intensities of solvent decomposition species in the C 1s spectrum, as well as the reduced intensity of LiF/MF₂ in the F 1s spectrum (Figure S20, Supporting Information).^[29] Furthermore, the lower intensity ratio of LiPF_xO_y to LiPF_6 in the P 2p spectrum of the ICEP-8 cathode (0.24) compared to that of the PVDF cathode (0.79) indicated the suppression of salt degradation. The thinner and uniform CEI layer in the ICEP-8 cathode is attributed to the uniform polymer coverage on the active material, which minimizes the exposure of active materials to the liquid electrolyte. In contrast, the non-uniform distribution of the PVDF binder led to localized binder deficiencies, exposing NCM811 to continuous parasitic side reactions with the electrolyte. This instability resulted in severe CEI growth, persistent electrolyte decomposition, and cathode deterioration in the PVDF system.

The structural robustness of NCM811 cathodes with the ICEP-8 binders was investigated using synchrotron-based nano-computed tomography (nano-CT) to examine morphological evolution and the formation of internal particle cracks at high resolution.^[5b,30] Unlike conventional imaging techniques, nano-CT enables 3D visualization of subsurface cracks and structural distortions that are otherwise undetectable, offering a field of view of 76 μm with a pixel resolution of 44 nm. The 3D reconstructed images revealed stark contrasts between the two cathodes: the PVDF cathode exhibited severe particle pulverization with microcracks (white arrows), whereas the ICEP-8 cathode retained intact particle morphology and structural integrity (Figure 5a,b). A similar trend was observed in XY-plane transmission X-ray microscopy (TXM) images, where extensive intergranular cracking was detected in the PVDF cathode, while the ICEP-8 cathode exhibited no noticeable cracks. These findings were further corroborated by cross-sectional SEM imaging, which showed pronounced internal cracks within the NCM811 particles of the PVDF cathode (Figure S21, Supporting Information). The observed intergranular fractures in the PVDF cathode are attributed to inhomogeneous lithiation and delithiation, caused by poor binder coverage and insufficient Li-ion transport pathways. In contrast, the ICEP-8 binder provided uniform particle coverage, facilitating homogeneous Li-ion diffusion and preventing localized stress buildup, thereby suppressing crack formation.

To further probe the atomic-scale effects of the binder on structural stability, scanning transmission electron microscopy (STEM) coupled with fast Fourier transform (FFT) and electron energy loss spectroscopy (EELS) analyses were conducted. These

techniques provided direct evidence that the ICEP-8 binder mitigates phase transitions in NCM811 particles. The transformation of the layered structure into a rock-salt phase is a well-known degradation mechanism that hinders Li-ion diffusion and reduces Li-storage capacity. Notably, the ICEP-8 cathode exhibited a drastically thinner phase transformation layer (≈ 3.1 nm) at site A compared to the PVDF cathode, which developed a significantly thicker transformation region (≈ 11.3 nm) (Figure 5c,d). EELS spectral analysis further confirmed the stabilization effect of the ICEP-8 binder on TM oxidation states (Figure 5e,f). The energy loss difference (ΔE) in the O K-edge spectra, defined as the gap between the pre-edge peak (≈ 528 eV) and the main-edge peak (≈ 540 eV), revealed that the ICEP-8 cathode maintained higher ΔE values, indicating a suppressed reduction of Ni^{4+} to lower valence states (Ni^{2+} and Ni^{3+}) on the particle surface.^[31] Furthermore, the O K pre-edge peak appeared at a lower depth in the ICEP-8 cathode, suggesting lower oxygen loss and improved surface stability. Additionally, the Ni L_3/L_2 ratio exhibited a steeper decline from surface to bulk in the ICEP-8 cathode, demonstrating its ability to suppress Ni^{2+} migration into Li slabs, causing rock-salt phase thickening and structural degradation. These findings indicate that the ICEP-8 binder plays a crucial role in preserving the layered structure of NCM811, effectively mitigating phase transitions and ensuring long-term electrochemical stability.

The enhanced structural integrity of ICEP-8 cathodes was further corroborated by differential capacity (dQ/dV) analyses of cycling data presented in Figure 3a. The dQ/dV curves of the PVDF cathode exhibited significant peak shifts, particularly around 3.65 V, corresponding to the original hexagonal to monoclinic (H1-to-M) phase transition. Such peak shifts reflect severe polarization caused by inhibited Li-ion transport and structural deterioration within the cathode (Figure S22, Supporting Information).^[11] In contrast, the ICEP-8 cathode demonstrated minimal peak shifts throughout extended cycling, highlighting superior particle stability imparted by the ICEP-8 binder. Collectively, these results establish the multifunctional ICEP-8 binder as a critical element for preserving structural and interfacial stability, effectively suppressing TM dissolution, minimizing side reactions, and mitigating phase transitions, thereby significantly enhancing the long-term electrochemical performance of ultrahigh mass-loading NCM811 cathodes.

2.5. Electrochemical Performance of Ultrahigh Loading Cathodes and High-Energy-Density Li Metal Pouch Cells

Building on the superior chemomechanical properties of the ICEP-8 binder, high-mass-loading cathodes were fabricated to demonstrate their applicability in high-energy-density Li batteries. ICEP-8 cathodes were prepared with varying areal mass loadings of 18.8 mg cm^{-2} (3.4 mAh cm^{-2}), 36.9 mg cm^{-2} (6.6 mAh cm^{-2}), 45.3 mg cm^{-2} (8.8 mAh cm^{-2}), 52.3 mg cm^{-2} (10.5 mAh cm^{-2}), and 62.4 mg cm^{-2} (12.5 mAh cm^{-2}). Cross-sectional SEM images of the as-prepared high-loading revealed a compact and uniform packing layer with no visible voids or cracks even at the highest mass loading of 62.4 mg cm^{-2} (Figure S23, Supporting Information). This well-preserved electrode morphology suggests that the ICEP-8 binder effectively maintains structural

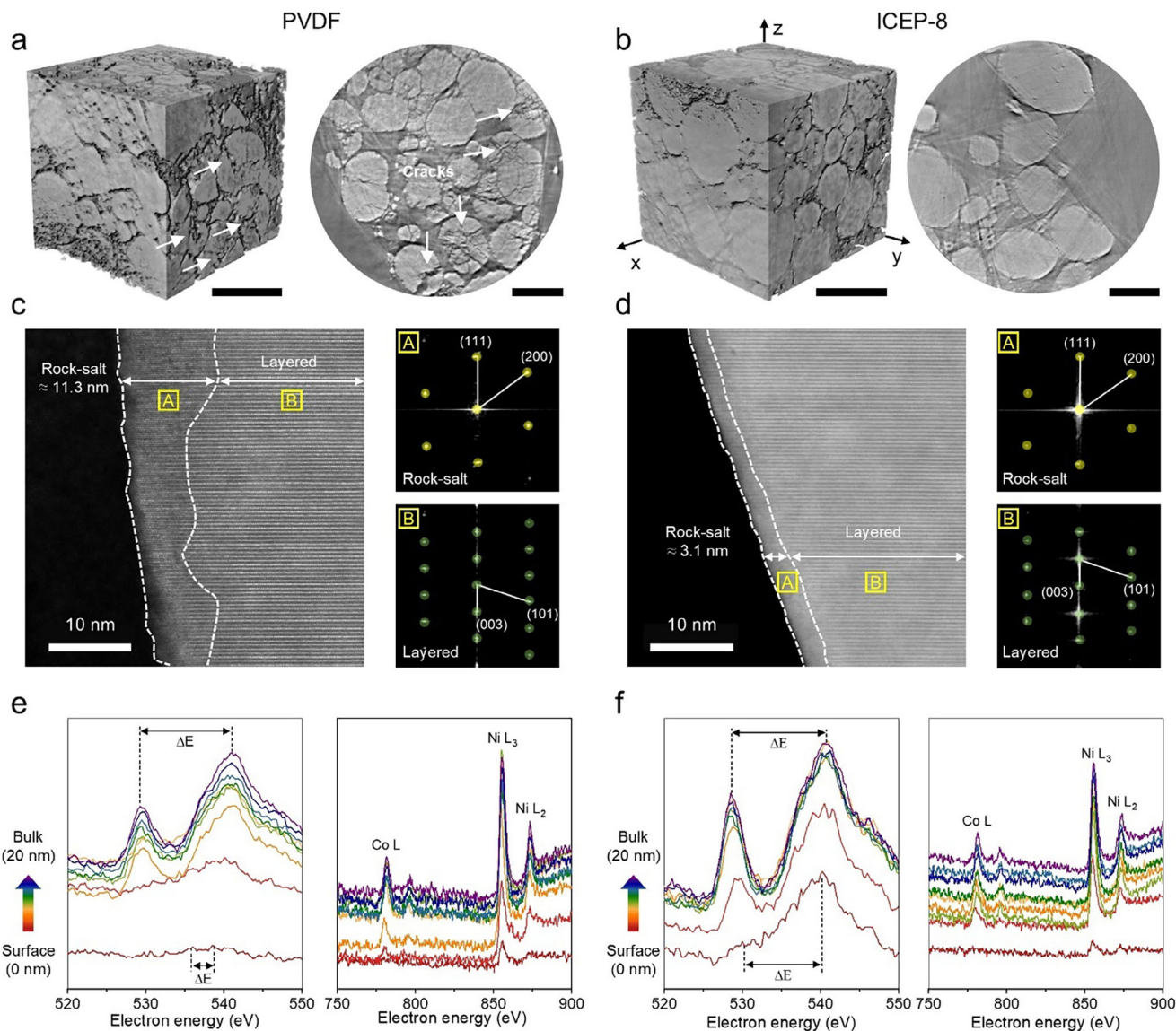


Figure 5. Structural stability analysis of cycled NCM811 cathodes. Synchrotron-based X-ray nano-computed tomography images of cycled cathodes with a) PVDF and b) ICEP-8 binders (Scale bar = 10 μm). STEM images with corresponding FFT patterns for NCM811 cathodes with c) PVDF and d) ICEP-8 binders. EELS spectra (O K-, Co L-, Ni L₃-, and L₂-edges) of cycled NCM811 cathodes with e) PVDF and f) ICEP-8 binders.

cohesion, even under high mass loading conditions. To evaluate full cell performance, these high-areal-capacity cathodes were paired with a fixed Li metal anode (100 μm $\approx$ 20 mAh cm^{-2}), resulting in Li metal full cells with N/P ratios ranging from 1.6 to 5.9. The robust adhesion properties and enhanced ion-conduction pathways provided by the ICEP-8 binder facilitated stable and proportional increases in the areal capacity as the cathode mass loading increased (Figure 6a). Notably, the high mass loading achieved with the ICEP-8 binder far exceeded that of previously reported cathode binders, highlighting its exceptional capability for thick electrodes (Figure 6b; Table S2, Supporting Information). Furthermore, the binder content of the cathode was reduced to 1 wt.% and investigated performance at areal loadings of 19.5, 37.3, and 60.9 mg cm^{-2} to evaluate industrial feasibility (Figure S24, Supporting Information). All electrodes exhib-

ited stable voltage profiles in the formation cycle and delivered areal capacities of 3.91, 7.47, and 12.3 mAh cm^{-2} , respectively, with no compromise in specific capacity. During prolonged cycling at 0.1 C/0.1 C, all three cathodes with minimal binder content retained stable cycling behavior, confirming the outstanding effectiveness of ICEP-8 even at reduced concentrations.

The ultimate goal of high-areal-capacity electrodes is to achieve high-energy-density full cells without compromising specific capacity. While previous studies often overlook the trade-off between mass loading and energy density, the ICEP-8 binder enables a consistent increase in gravimetric energy density with increasing cathode mass loading, reaching a plateau of 424.4 Wh kg^{-1} at 62.4 mg cm^{-2} (Figure 6c; Calculation details in Table S3, Supporting Information; excluding package materials). In contrast, the cells assembled with PVDF cathodes exhibited

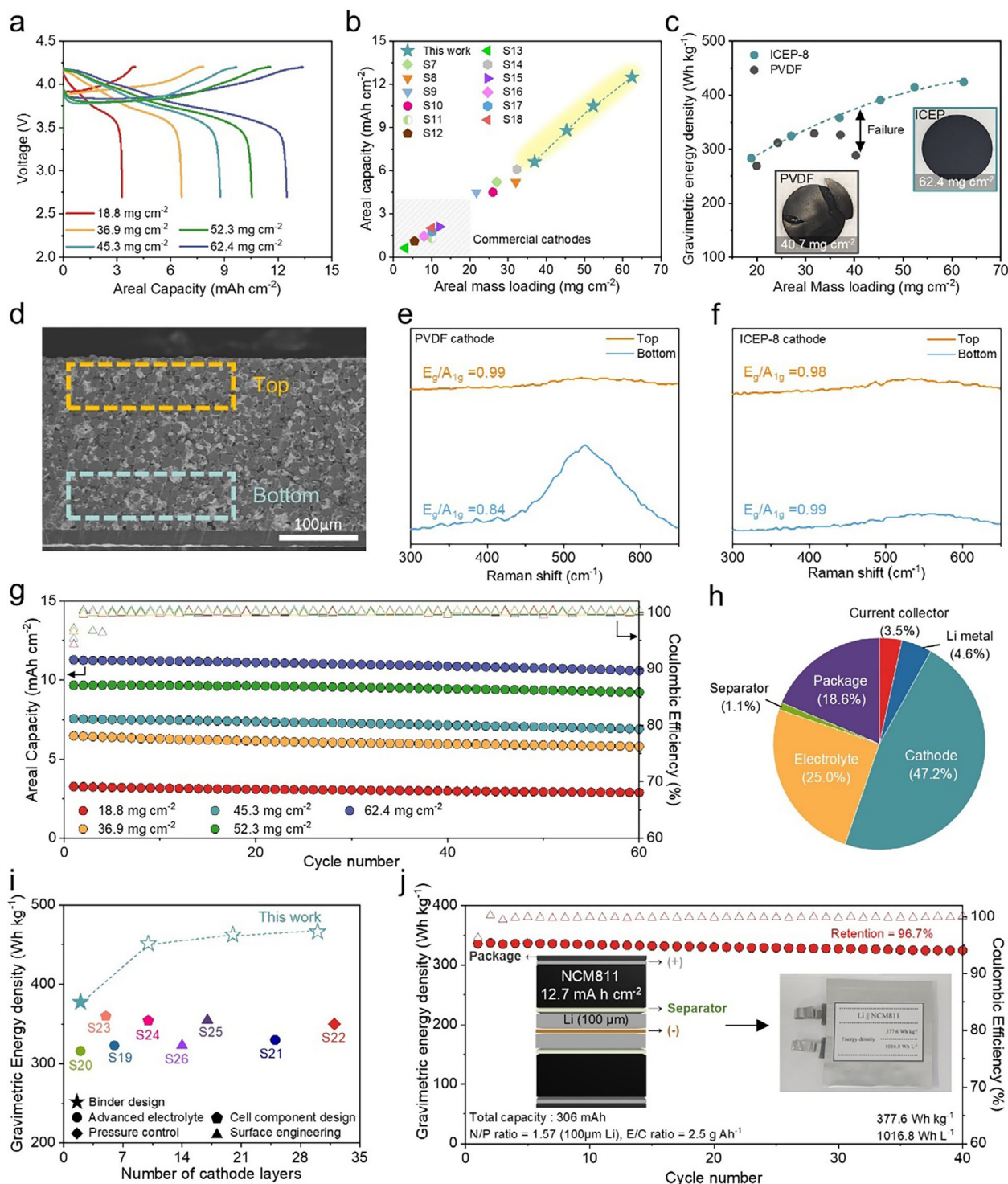


Figure 6. Electrochemical performance and energy density of high-mass-loading ICEP-8 cathodes. a) Galvanostatic charge/discharge profiles at increasing areal capacities of ICEP-8 cathodes. b) Comparison of areal capacity as a function of mass loading between ICEP-8 cathodes (this work) and previously reported cathode binders. c) Gravimetric energy density of coin-type cells versus areal mass loading for ICEP-8 and PVDF cathodes (Inset: Photographs of PVDF and ICEP-8 cathodes). d) Cross-sectional SEM image of fully charged ICEP-8 cathode (4.3 V) for Raman spectroscopy analysis. Raman spectra and E_g/A_{1g} intensity ratios at top and bottom regions of e) PVDF and f) ICEP-8 cathodes. g) Cycling stability of Li metal full cells with varying mass loadings of ICEP-8 cathodes. h) A pie chart of the weight distributions of components in a double-stacked Li metal pouch-type full cell. i) Comparative plot of gravimetric energy density against cathode stack number in Li metal pouch-type full cells. j) Energy density retention of double-stacked Li metal pouch cell.

a decline in gravimetric energy density beyond 31.7 mg cm^{-2} , primarily due to severe cracking and electrode delamination at a mass loading of 40.7 mg cm^{-2} . These failures highlight the inadequate chemomechanical properties of PVDF binders for high-mass-loading applications. The fundamental difference in energy density trends between ICEP-8 and PVDF cathodes can be attributed to two key factors: the elasticity of the ICEP-8 binder and its ability to ensure through-thickness redox uniformity by a stable ionic-conduction pathway, which governs capacity realization.

To further elucidate the redox uniformity in the high-mass-loading electrodes, confocal Raman spectroscopy was employed to analyze the cathodes before and after charging, at different locations within the electrodes, specifically at the “top” and “bottom” regions, based on their distance from the current collector (Figure 6d). The Raman spectra of metal oxide cathodes, such as NCM811, typically exhibit two characteristic modes: the A_{1g} mode (530 cm^{-1}), associated with the stretching of the metal-oxygen complex along the *c*-axis, and the E_g mode (440 cm^{-1}), which corresponds to the bending of the metal-oxygen-metal complex in the *a/b* axis direction. These modes are directly related to the SOC of the material, with the intensity ratio of E_g/A_{1g} generally increasing as the SOC increases. The intensity ratio of E_g/A_{1g} was calculated at different locations within the cathodes at varying charge states. In the pristine state, the E_g/A_{1g} ratio of the PVDF cathode was 0.83 (Figure S25, Supporting Information). At the fully charged state, the PVDF cathode exhibited a higher ratio of 0.99 at the top but a similar ratio of 0.84 at the bottom compared to the pristine state. These results indicate the NCM811 material near the current collector was less efficient in de-lithiation, attributed to the uneven Li-ion pathways within the cathode which leads to limited Li-ion diffusion in the bottom region (Figure 6e). In contrast, the ICEP-8 cathode maintained a consistent E_g/A_{1g} ratio (0.99 at the bottom and 0.98 at the top) across the entire cross-section, demonstrating uniform redox behavior throughout the cathode (Figure 6f). This observation reconfirmed the through-thickness redox uniformity facilitated by the ICEP-8 binder. The cycling performance of full cells with ICEP-8 cathodes at varying N/P ratios was evaluated at a 0.1/0.1 C current density (Figure 6g). All cells exhibited stable cycling performance throughout 60 cycles, with 52.3 mg cm^{-2} and 62.4 mg cm^{-2} demonstrating 96.3% and 94.6% capacity retention, respectively. In contrast, the PVDF cathode, despite a relatively lower mass loading of 37.3 mg cm^{-2} , displayed severe capacity fade, retaining only 54.7% of its initial capacity after 60 cycles (Figure S26, Supporting Information). These results indicate that the excellent chemomechanical properties of the ICEP-8 binder enable to form stable CEI layer and enhance charge transfer kinetics even in ultrahigh mass-loading configurations, contributing to outstanding cycling stability.

For commercial applicability, a double-stacked pouch-type full cell was fabricated, incorporating ultrahigh loading ICEP-8 cathodes (62.5 mg cm^{-2} , 12.7 mAh cm^{-2}) paired with Li metal anodes ($100 \text{ } \mu\text{m}$, N/P ratio of 1.57) and a controlled E/C ratio of 2.5 g Ah^{-1} . The pouch cell delivered a capacity of 304 mAh, achieving a remarkable energy density of $377.6 \text{ Wh kg}_{\text{cell}}^{-1}/1016.8 \text{ Wh L}_{\text{cell}}^{-1}$ (Figure S27 and Table S4, Supporting Information). The high-energy-density of this pouch-type full cell can be attributed to the high-areal-capacity of the ICEP-8 cathodes, with the reported energy density representing the total en-

ergy delivered from the entire pouch cell, including both active and inactive components (Figure 6h). To further enhance the energy density, multicell stacking configurations were considered, which would reduce the weight fraction of packaging materials, thereby improving the overall energy-to-weight ratio. We calculated the estimated gravimetric energy density for the multicell stacking configurations, assuming the same mass loading (62.5 mg cm^{-2}) for each cathode sheet as used in the double-stacked pouch-type full cell (Table S5, Supporting Information). Notably, even with just a double-stacked pouch cell configuration (two cathode sheets) fabricated in this work, the gravimetric energy density of the pouch cell outperformed previously reported Li metal pouch-type full cells employing various strategies with multicell stacking configurations (Figure 6i and Table S6, Supporting Information). As the number of cathode layers increased to 10, 20, and 30, the estimated gravimetric energy density further increased to 451.3, 462.3, and $466.1 \text{ Wh kg}_{\text{cell}}^{-1}$, respectively, driven by the high-areal-capacity of the cathodes and the reduced portion of packaging materials. Finally, the energy density retention of the double-stacked pouch cell with ICEP-8 cathodes was evaluated at 0.1 C/0.1 C, demonstrating exceptional energy density retention of 96.7% after 40 cycles (Figure 6j). These results collectively highlight the potential of the multifunctional ICEP binder as the most effective approach for increasing the energy density of lithium batteries, enabling both superior performance and enhanced energy density. However, a sudden capacity drop was observed beyond 40 cycles, which is attributed to Li metal degradation under the combined conditions of high areal capacity ($\approx 12.7 \text{ mAh cm}^{-2}$) and lean electrolyte ($E/C = 2.5 \text{ g Ah}^{-1}$). This instability is exacerbated by the use of conventional carbonate electrolytes, which are known to form unstable, organic-rich SEI layers on Li metal, leading to dendritic growth and rapid electrolyte depletion.^[32] To confirm this, the cycled pouch cell was disassembled, and the ICEP-8 cathodes were carefully rinsed, dried, and reassembled into fresh coin-type full cells using pristine Li metal and electrolyte. The reassembled cells showed excellent capacity retention, confirming that the performance decline was due to Li degradation rather than cathode instability (Figure S28, Supporting Information). While increasing N/P or E/C ratios could mitigate this degradation, it would reduce overall energy density. Therefore, the development of advanced electrolytes, particularly high-voltage-compatible ether systems, is critical to simultaneously extend cycle life and preserve energy density in Li metal pouch cells.^[33]

3. Conclusion

In summary, the ionically conductive elastic polymer (ICEP) binder was developed as a multifunctional binder to enable the fabrication of ultrahigh mass-loading Ni-rich cathodes while simultaneously enhancing ion/electron conduction to realize high-energy-density Li batteries. The ICEP binder integrates mechanical elasticity, strong adhesion, and ionic conductivity through its diverse functional groups, addressing longstanding challenges in the fabrication of high-areal-capacity electrodes. Strong hydrogen bonding interactions between the ICEP binder and NCM811 particles ensure uniform electrode morphology, facilitating the formation of a stable and homogeneous cathode-electrolyte interphase (CEI). This mitigates electrolyte-induced

side reactions, suppresses transition metal dissolution, and enhances long-term electrochemical stability. Beyond interfacial stabilization, the ICEP binder significantly improves Li-ion diffusivity and reduces interphase/charge transfer resistance, thereby promoting faster reaction kinetics and enhanced electrochemical performance in ultrathick electrodes. Moreover, the elasticity of the ICEP binder effectively accommodates solvent-drying-induced stress, preventing cracking and preserving electrode integrity. As a result, high-mass-loading electrodes (62.4 mg cm^{-2} , 12.5 mAh cm^{-2}) were successfully fabricated, demonstrating 94.6% capacity retention in coin-type Li metal full cells. Furthermore, its implementation in double-stacked Li metal pouch cells achieved an energy density of $377.6 \text{ Wh kg}_{\text{cell}}^{-1}$ and $1016.8 \text{ Wh L}_{\text{cell}}^{-1}$ (including package materials), among the highest values reported for Li metal batteries. These findings establish ICEP as a highly effective multifunctional binder strategy for next-generation high-energy-density batteries, providing a scalable and commercially viable solution for ultrahigh-loading cathodes.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

cathode binders, high-energy-density lithium batteries, ionically conductive polymers, nickel-rich layered cathodes, ultrahigh mass loadings

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